

house. If it was expected to take two years, and three years later you are asking questions, do not expect the project to be completed any time soon. If wars last on average six months, and your conflict has been going on for two years, expect another few years of problems. The Arab-Israeli conflict is sixty years old, and counting—yet it was considered “a simple problem” sixty years ago. (Always remember that, in a modern environment, wars last longer and kill more people than is typically planned.) Another example: Say that you send your favorite author a letter, knowing that he is busy and has a two-week turnaround. If three weeks later your mailbox is still empty, do not expect the letter to come tomorrow—it will take on average another three weeks. If three months later you still have nothing, you will have to expect to wait another year. Each day will bring you closer to your death but further from the receipt of the letter.

This subtle but extremely consequential property of scalable randomness is unusually counterintuitive. We misunderstand the logic of large deviations from the norm.

I will get deeper into these properties of scalable randomness in Part Three. But let us say for now that they are central to our misunderstanding of the business of prediction.

DON'T CROSS A RIVER IF IT IS (ON AVERAGE) FOUR FEET DEEP

Corporate and government projections have an additional easy-to-spot flaw: they do not attach a *possible error rate* to their scenarios. Even in the absence of Black Swans this omission would be a mistake.

I once gave a talk to policy wonks at the Woodrow Wilson Center in Washington, D.C., challenging them to be aware of our weaknesses in seeing ahead.

The attendees were tame and silent. What I was telling them was against everything they believed and stood for; I had gotten carried away with my aggressive message, but they looked thoughtful, compared to the testosterone-charged characters one encounters in business. I felt guilty for my aggressive stance. Few asked questions. The person who organized the talk and invited me must have been pulling a joke on his colleagues. I was like an aggressive atheist making his case in front of a synod of cardinals, while dispensing with the usual formulaic euphemisms.

Yet some members of the audience were sympathetic to the message. One anonymous person (he is employed by a governmental agency) explained to me privately after the talk that in January 2004 his department was forecasting the price of oil for twenty-five years later at \$27 a barrel, slightly higher than what it was at the time. Six months later, around June 2004, after